

ST LAWRENCE, NL, Canada

WMO: 711100

Lat: 46.920N

Lon: 55.380W

Elev: 49

StdP: 100.74

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.6	-10.9	-15.8	0.9	-12.0	-14.2	1.1	-10.3	22.5	-4.3	21.1	-4.9	5.8	310	0.795

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	23.1	19.1	21.6	18.2	20.2	17.6	20.3	21.8	19.1	20.5	18.2	19.5	5.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	21.0	18.6	13.5	19.8	17.7	12.7	18.9	58.6	21.9	54.7	20.5	51.7	19.5	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 18.1	15.7	13.5		DB	-15.6	26.0	1.7	2.0	-16.8	27.5	-17.8	28.6	-18.7	29.8	-19.9	31.2
(5)				WB	-15.8	21.2	1.7	1.6	-17.0	22.3	-18.0	23.3	-18.9	24.2	-20.2	25.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-2.9	-3.5	-1.4	2.4	6.7	10.7	15.3	16.7	13.4	8.6	4.3	-0.1
	DBStd	7.54	3.76	4.01	3.54	2.71	2.54	2.69	2.48	2.34	3.01	3.26	3.41	3.90
	HDD10.0	2045	398	378	355	228	106	22	0	0	6	67	172	313
	HDD18.3	4546	657	611	613	478	359	228	97	61	149	302	419	572
	CDD10.0	556	0	0	0	0	5	44	165	207	109	24	2	0
	CDD18.3	16	0	0	0	0	0	0	4	9	2	0	0	0
	CDH23.3	43	0	0	0	0	1	3	14	17	9	0	0	0
	CDH26.7	4	0	0	0	0	0	0	1	1	2	0	0	0
(14) Wind	WSAvg	6.2	7.7	7.9	7.2	6.5	5.5	5.0	4.3	4.7	5.3	6.1	6.7	7.4
(15) Precipitation	PrecAvg	1603	141	133	129	124	118	142	110	130	147	154	148	137
	PrecMax	1987	216	246	208	238	203	348	198	410	253	273	259	271
	PrecMin	1218	59	44	73	57	58	50	46	53	56	41	81	76
	PrecStd	191	35	57	34	55	33	67	41	75	45	69	40	41
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	7.2	8.5	12.1	19.2	22.5	25.0	25.2	23.7	18.6	13.8	10.6
		MCWB	7.6	6.7	7.2	8.0	13.0	16.4	19.6	20.1	19.2	16.8	13.3	10.5
	2%	DB	5.3	5.1	6.0	9.4	15.3	19.3	22.8	23.3	20.5	16.6	12.1	8.8
		MCWB	5.1	4.8	4.7	6.9	10.5	14.7	18.5	19.6	17.8	15.1	11.7	8.4
	5%	DB	3.8	3.6	4.4	7.8	13.1	17.3	21.1	21.9	19.1	15.1	11.0	7.2
		MCWB	3.5	3.2	3.5	6.0	9.3	13.5	17.6	18.9	17.1	13.9	10.6	6.8
	10%	DB	2.4	2.0	3.2	6.5	11.3	15.5	19.5	20.6	18.0	13.8	9.7	5.5
		MCWB	2.0	1.5	2.4	5.1	8.6	12.6	17.0	18.2	16.4	12.6	9.1	5.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	6.9	7.5	9.2	14.3	17.9	21.9	22.8	20.6	17.8	13.1	10.6
		MCDB	7.8	7.0	8.3	10.5	17.5	20.3	23.5	23.8	22.7	17.9	13.2	10.7
	2%	WB	5.1	4.8	5.1	7.6	11.7	15.7	19.7	20.7	18.8	15.6	11.7	8.5
		MCDB	5.2	5.0	5.6	8.6	13.8	18.2	21.4	22.0	20.0	16.3	11.9	8.7
	5%	WB	3.6	3.2	3.7	6.4	10.2	14.3	18.4	19.6	17.8	14.2	10.6	6.8
		MCDB	3.7	3.5	4.2	7.4	12.0	16.3	19.9	21.0	18.6	14.9	10.9	7.1
	10%	WB	2.1	1.6	2.4	5.3	9.0	13.1	17.5	18.7	16.8	12.7	9.2	5.3
		MCDB	2.3	2.0	3.0	6.3	10.9	14.9	18.8	19.9	17.7	13.5	9.6	5.6
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.1	6.2	6.7	7.9	7.7	6.8	6.6	7.4	6.7	6.0	5.7
		MCDBR	7.6	7.4	7.1	8.6	11.4	11.2	9.7	8.6	8.0	7.4	7.1	7.2
	5% WB	MCWBR	8.0	7.4	6.2	6.5	7.3	7.2	6.3	5.7	5.9	6.1	7.0	7.4
		MCWBR	7.6	7.3	6.9	7.8	9.3	9.2	7.5	6.8	6.9	6.8	6.8	7.2
(40) Clear-Sky Solar Irradiance	taub		0.322	0.321	0.335	0.369	0.377	0.400	0.430	0.403	0.370	0.354	0.327	0.314
		taud	2.349	2.426	2.428	2.358	2.395	2.355	2.318	2.398	2.468	2.456	2.460	2.383
	Ebn at Noon		754	850	891	886	888	866	833	843	842	798	751	720
		Edh at Noon	74	84	97	115	115	120	123	109	93	81	67	64
(44) All-Sky Solar Radiation	RadAvg		0.98	1.72	2.92	4.20	5.04	5.14	4.88	4.36	3.59	2.23	1.24	0.83
	RadStd		0.08	0.16	0.17	0.36	0.39	0.53	0.34	0.28	0.23	0.16	0.16	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page